REACH: an accessible and regionally portable air quality model

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1. REACH Model Description

REACH is a reduced-complexity model with a Gaussian-plume framework and can be used to examine air pollution policies. The model is designed to be location-agnostic by allowing for adjustment of input data and geography information. Every location in the model (sources and receptors) can be designated by their population-weighted/geometric centroids or any user-designated coordinates.

There are three main outputs of the REACH model:

1.1 Ambient PM2.5 concentrations

The model predicts ambient PM2.5 concentrations from emissions of primary PM2.5, SO2, NOx, NH3 and VOCs.

1.2 Source-receptor matrices

A source is a point location of emissions and a receptor is a location upwind of an emission source. A source-receptor vector is the change in annual-average PM2.5 concentrations downwind from a source location from a 1 tonne change in emissions of a PM2.5 precursor. The source-receptor matrices represent the following relationships:

change in emissions –> change in concentrations (ug/m3/tonne)

- primary PM2.5 –> primary PM2.5
- SO2 –> sulfate
- NOx –> total nitrate (gas + particle)
- NH3 –> total ammonium (gas + particle)
- VOCs –> SOA

1.3 Marginal social costs

Source location-specific aggregate PM2.5 health impacts from emitting a tonne of a PM2.5 precursor (mortalities/tonne or $/tonne).

For detailed model description, please see the main paper: Akindele, Medinat, Peter J. Adams, Nicholas Z. Muller. "REACH: an accessible and globally portable air quality model for policy and health impact assessment." In preparation.

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The following sections outline description of the main folders, files, and scripts and provides high-level instructions to run the model.

For detailed instructions, please see the REACH user guide on Github.

Link to Zenodo for additional files: https://zenodo.org/records/20546830

2. Description of folders

RCM_workshop

This directory contains valuable resources and practice problems for those interested in learning about Reduced-Complexity Models (RCMs). It includes a presentation on RCMs with an accompanying audio recording by Dr. Peter Adams at Carnegie Mellon University. The folder also features background materials and application papers related to RCMs. Additionally, five practice problems are provided to help reinforce key concepts.

REACH_MAIN

This directory contains the model scripts to generate the source-receptor matrices. Version A and Version B scripts are similar, except that Version A is suitable for emission inventories where stack information for point sources is available. For less detailed emission inventories, Version B is more suitable.

In Version_A and Version_B are the following folders:

- model:
  - domain_inputs.R: Edit to specify inputs for the domain (list of emission sources and receptor locations, sub divisions (grids within each source location), and the name specified for the location identifier)
  - input_info.R: Input variables and functions that are used for the model_script
  - model_script.R: Script to generate source-receptor matrix
- source_receptor: source-receptor scripts for each pollutant and height that link to the scripts in the model folder
- dispersion: contains the Gaussian plume equation and relevant loss processes (deposition and/or chemistry) for each pollutant

REACH_SA

Directory for REACH-Southern Africa (SA) with input data, post-processing scripts, and results. For REACH-SA, the Version B scripts in REACH_MAIN were used.

List of folders:

- inputs
  - emissions: emission files for each pollutant and height
  - met_data: meteorology files for each source location. For all files, download met_data_SA.zip at the Zenodo link.
- results
  - concentrations: baseline PM2.5 concentrations
  - matrices: source-receptor matrices for each pollutant and height. Each row represents a source location and each column represents a receptor location. The units are ug/m3/tonne.
  - social_costs: marginal social costs (mortalities per tonne) for each source location
- scripts
  - concentrations: post-processing scripts to get baseline PM2.5 concentrations
  - GBD_mortality: folder with scripts to estimate marginal social costs. Please read GBD_method_MSC.docx prior to preparing input data and running scripts.
  - sectors: scripts to compute sector-specific PM2.5 concentrations and mortality using the zero-out method

REACH_US

Directory for REACH-United States (US) with input data, post-processing scripts, and results. For REACH-US, the Version A scripts in REACH_MAIN were used.

List of folders:

- inputs
  - Sub-folders:
    - emissions: emission files for each pollutant and height
    - met_data: meteorology files for each source location. For all files, download met_data_US.zip at the Zenodo link.
  - Files:
    - input_info.txt: detailed information about each file in this folder
- results
  - Files:
    - output_info.txt: detailed information about each file in this folder (2017 National Emissions Inventory (NEI) results)
    - concentrations: baseline PM2.5 concentrations
    - matrices: source-receptor matrices of each pollutant and height. Each row represents a source location and each column represents a receptor location. The units are ug/m3/tonne. For all files, download matrices_US_2017.zip at the Zenodo link and add to this folder.
    - social_costs: marginal social costs ($/tonne) for each source location in the column named 'marginal_damage'
- additional_outputs: same sub-folders as the results folder, except for the 2005 NEI year. The S-R matrices are located at the Zenodo link (see matrices_US_2005.zip).
- scripts
  - concentrations: post-processing scripts to get baseline PM2.5 concentrations
  - social_costs: scripts to get marginal social costs
- tracts
  - The folder containing the tract source-receptor data can be made available upon reasonable request.

3. Steps to prepare the source-receptor matrices

Follow these steps to run the scripts for the source-receptor matrices. The folder and file descriptions have been described above.

- Step 1: Download Rstudio
  - https://posit.co/products/open-source/rstudio/
- Step 2: Create a main directory for the REACH project
- Step 3: Create two sub-directories in the main directory:

    - Folder for the domain: Add an R project folder (.Rproj extension)
    - Copy over the REACH_Main folder: Use VERSION A or VERSION B scripts

- Step 4: Create a file specifying coordinates (lat, lon) of emission locations (sources) and receptor locations

- Step 5: Prepare emissions and meteorology data files There are scripts available for global datasets: ERA5 meteorology (used for REACH-US and REACH-SA), EDGARv6.1 emissions (used for REACH-SA), and GFED biomass burning emissions (used for REACH-SA). For more information, see the REACH user guide. If using other datasets, prepare emission files for each pollutant and source location. Designate annual emissions in units of metric tonnes.

- Step 6: Generate additional input data

    - Sub-grids (05_subgrids.R)

- Step 7: Open the R project folder in Rstudio

    - Open model_script.R and input.info.R in VERSION A/VERSION B
    - Edit the domain_input.R script to designate emission locations and receptor locations for the domain

- Step 8: To run the source-receptor script for each pollutant and height

    - In VERSION_A/source_receptor or VERSION B/source_receptor folder:
    - Run each SR_pollutant_height script

4. Preparation of input data

The 'Inputs' folder contains scripts to prepare emissions and meteorology data for REACH. The scripts were prepared for global EDGAR emissions and ERA5 meteorology datasets as the default option. These datasets can be used when local emissions and meteorology data are unavailable. For the entire directory with input and output files, download 'Inputs.zip' at the Zenodo link. There are also scripts for DACCIWA emissions and can be made available upon reasonable request.

5. Post-processing scripts

5.1 Baseline PM2.5 concentrations

- Open the REACH_US.Rproj folder in Rstudio
- Edit and run the 01_concentration.R script
- Edit and run the 02_partition.R script

Adjust inputs as neccessary for the domain of interest

5.2 Marginal social costs

There are two methods used for estimating marginal social costs. If using the Global burden of disease method for PM2.5-attributed mortalities, use scripts in the REACH_SA/GBD_mortality folder. Otherwise if using all-cause mortality rates and relative risks for the United States, use scripts in the REACH_US/social_costs folder.

6. Calibration coefficients

Develop calibration coefficients for each PM2.5 species or use REACH-US calibrations as an assumption. Please see REACH user guide for more details.

7. Evaluation of total PM2.5

Lastly, after calibrating each PM2.5 species, aggregate the concentrations of each calibrated PM2.5 species to determine total PM2.5 (01_partition.R). Compare to measurements of total PM2.5 and ensure that the mean bias is relatively low due to the calibration process.

8. License